

ECE 593

Assignment 1

P1: Report/ Test Plan

Step 1: Specifications

- Design Name: ece593_alu
- Inputs:
 - A and B** (16-bits wide)
 - clk**: Clock signal for synchronization
 - rst**: Active high reset signal for to initialize ALU
 - op_sel**: ALU operation selector
 - start_op**: ALU operation control signal to start the operation
- Outputs:
 - result**: 32-bit wide to hold the output w.r.t inputs
 - end_op**: Signal indicating operation completion

Step 2: Verification Plan

- Create a parameterized SystemVerilog design and testbench that instantiates the DUT of **ece593_alu**
- Define the input stimulus (required signal to be randomized) for A, B, clk, rst, op_sel, and start_op.
- Monitor and Validate outputs (result and end_op) using assertions and check (for future)

Step 3: Testing

- Apply multiple combinations of A, B, and op_sel to verify the correctness of the operations performed.
- Assert and de-assert rst to ensure the ALU initializes correctly
- Test the start_op and end_op signals to ensure proper operation start and end

Step 4: Simulation Execution

- Use Questasim Simulator to run the testbench and simulate the ALU's behavior under various conditions
- Collect waveforms and transcripts to check with specifications

Step 5: Functional Coverage

- Ensure all operations (defined by op_sel) are worked out.
- Test with different types of test cases, such as border values of both input operands
- Verify the ALU behavior under simultaneous input changes

References: <https://github.com/raysalemi/uvmprimer>

P4:

Multiplication is more complex than addition and subtraction at the hardware level. It involves multiple steps (shifts, additions, carry propagation) compared to the simpler operations of addition and subtraction.

3 in binary: 00000011 (8 bits)

5 in binary: 00000101 (8 bits)

The multiplier 5 can be written as a sum of powers of two:

$$5 = 4 + 1 = 2^2 + 2^0$$

Multiply the powers of 2 with 3:

$3 \times 2^2 = 3 \times 4 = 12$ Left shift 3 by 2 bits $\rightarrow$ 00001100

$3 \times 2^0 = 3$ Left shift 3 by 0 bits $\rightarrow$ 00000011.

Binary Addition:

```
00001100
+ 00000011
-----
00001111 = 15 Final Result
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```

Steps to compile the code:

1. vlog ece593_alu.sv
2. vlog ece593_alu_tb.sv
3. vopt ece593_alu_tb -o ece593_alu_opt
4. vsim ece593_alu_opt